

Geometry-based Methods in Vision

CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

Administrivia

- Assignment 2, Problem Set 2 out
 - *Due on Tue next week (PS2), and Tue the week after (A2)*
- Discussion on Projects next week Thu

Lecture 10: Two-view Reconstruction



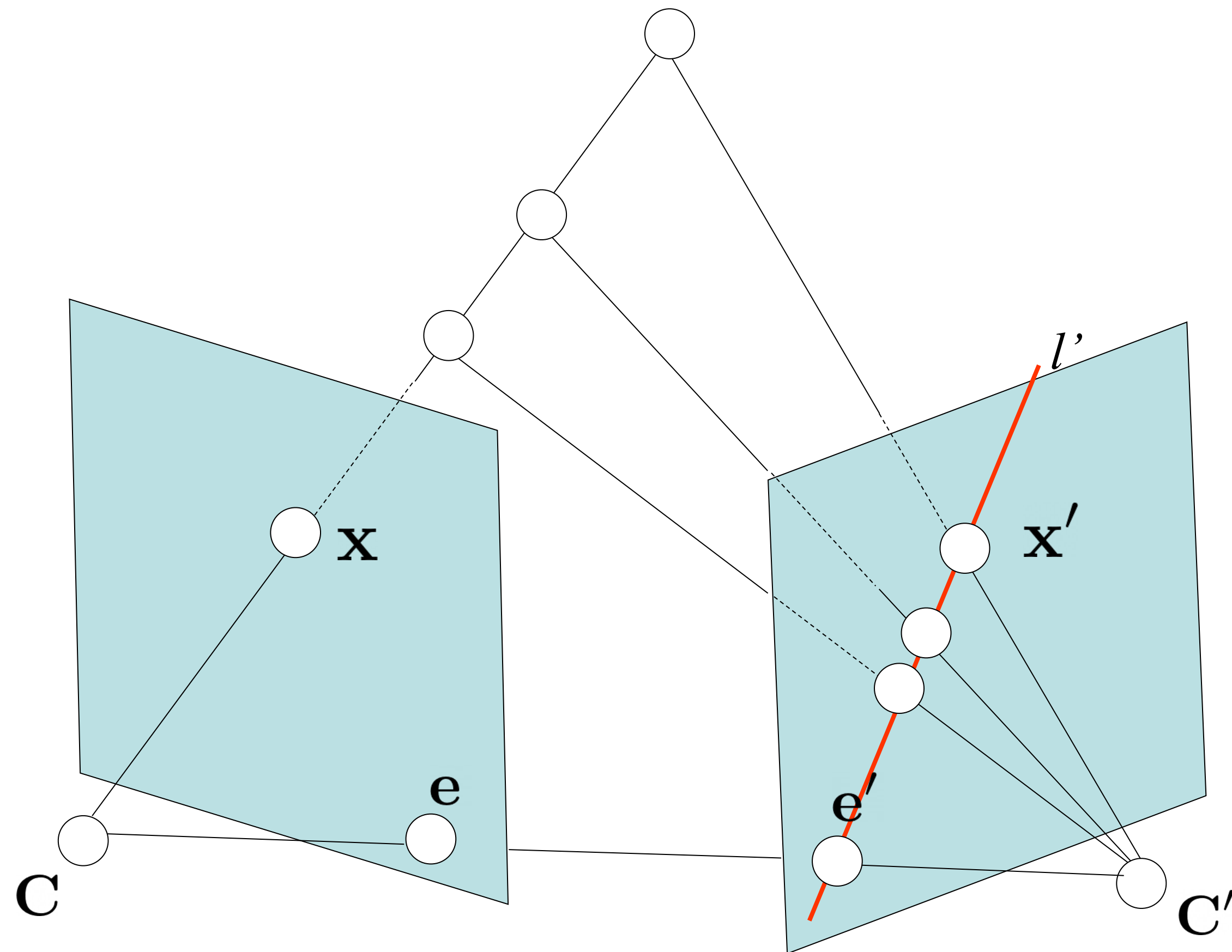
A reminder to ask questions!



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Review: Two-view Geometry



Point Correspondences:

$$\mathbf{x}'^\top F \mathbf{x} = 0$$

Epipoles:

$$\mathbf{e} = \mathbf{P} \mathbf{C}'$$

$$\mathbf{e}' = \mathbf{P}' \mathbf{C}$$

$$F \mathbf{e} = \mathbf{0}$$

$$\mathbf{e}'^\top F = \mathbf{0}^\top$$

Epipolar Lines:

$$\mathbf{l}' = F \mathbf{x}$$

$$\mathbf{l} = F^\top \mathbf{x}'$$

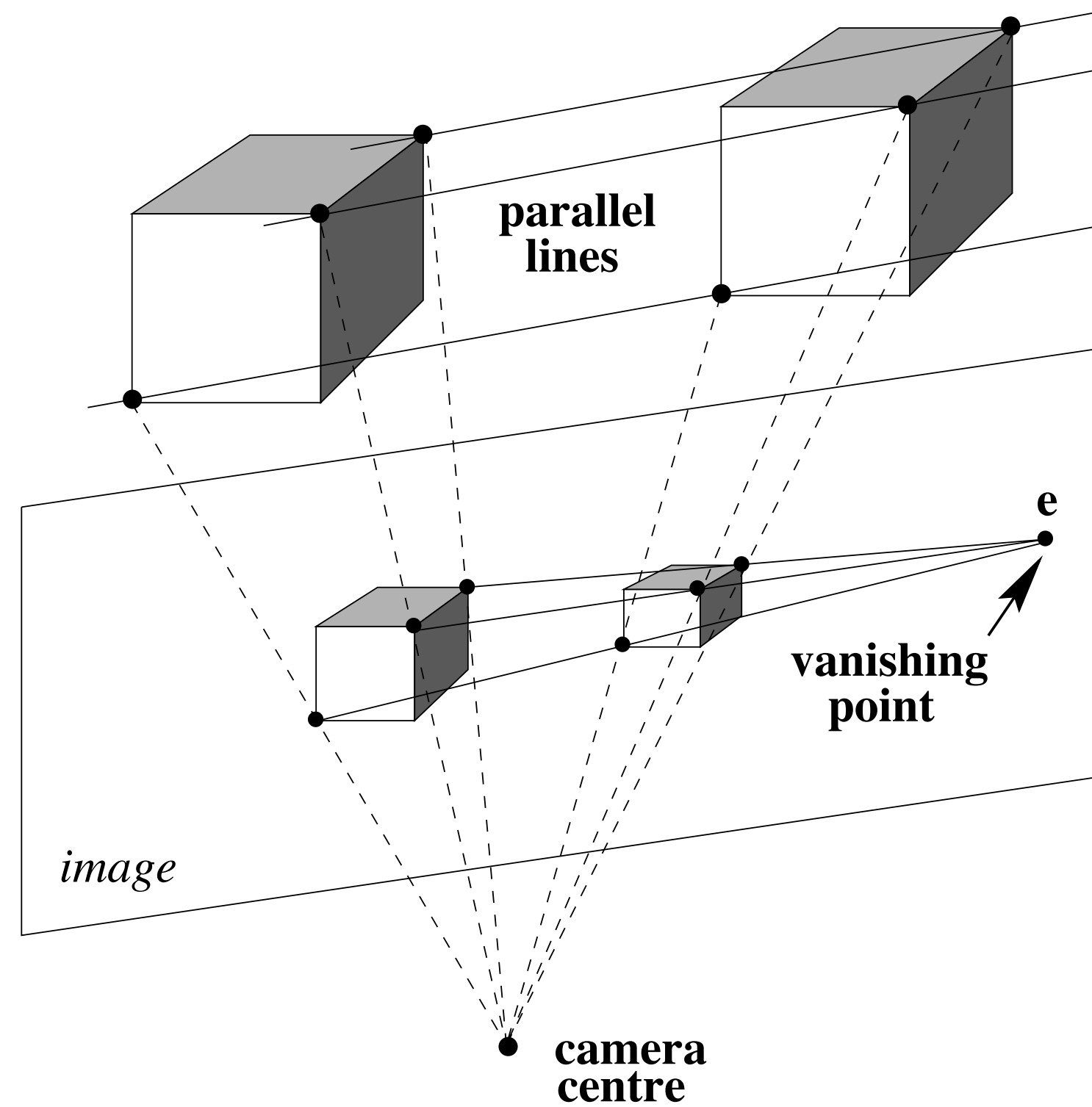
$$\mathbf{l}' = F[\mathbf{e}]_{\times} \mathbf{l}$$

$$\mathbf{l} = F^\top[\mathbf{e'}]_{\times} \mathbf{l}'$$

Rank and DoF:

$$F \equiv [\mathbf{f}_1 \quad \mathbf{f}_2 \quad \alpha \mathbf{f}_1 + \beta \mathbf{f}_2] \quad (\text{rank } 2, 7 \text{ DoF})$$

Special Case: Translation ($R=I$, $K=K'$)



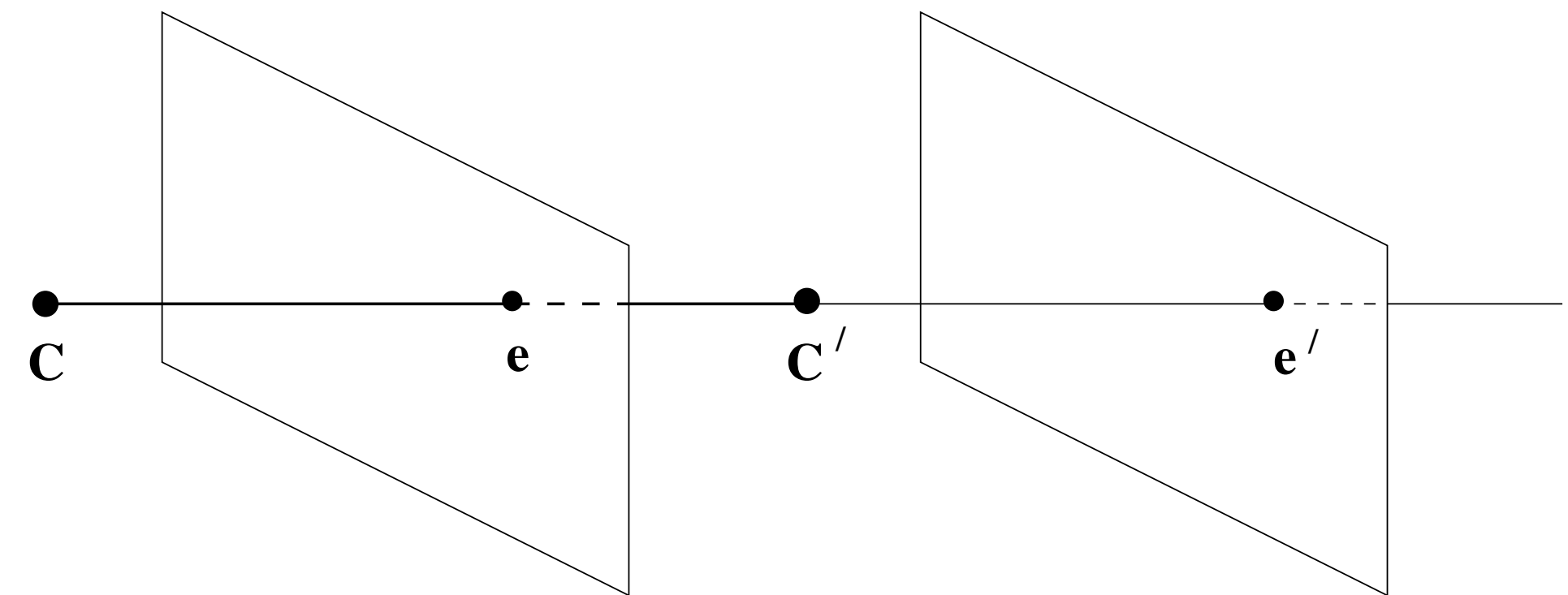
(similar to a static camera, and moving world)

$$F = [K't]_{\times} K' R K^{-1}$$

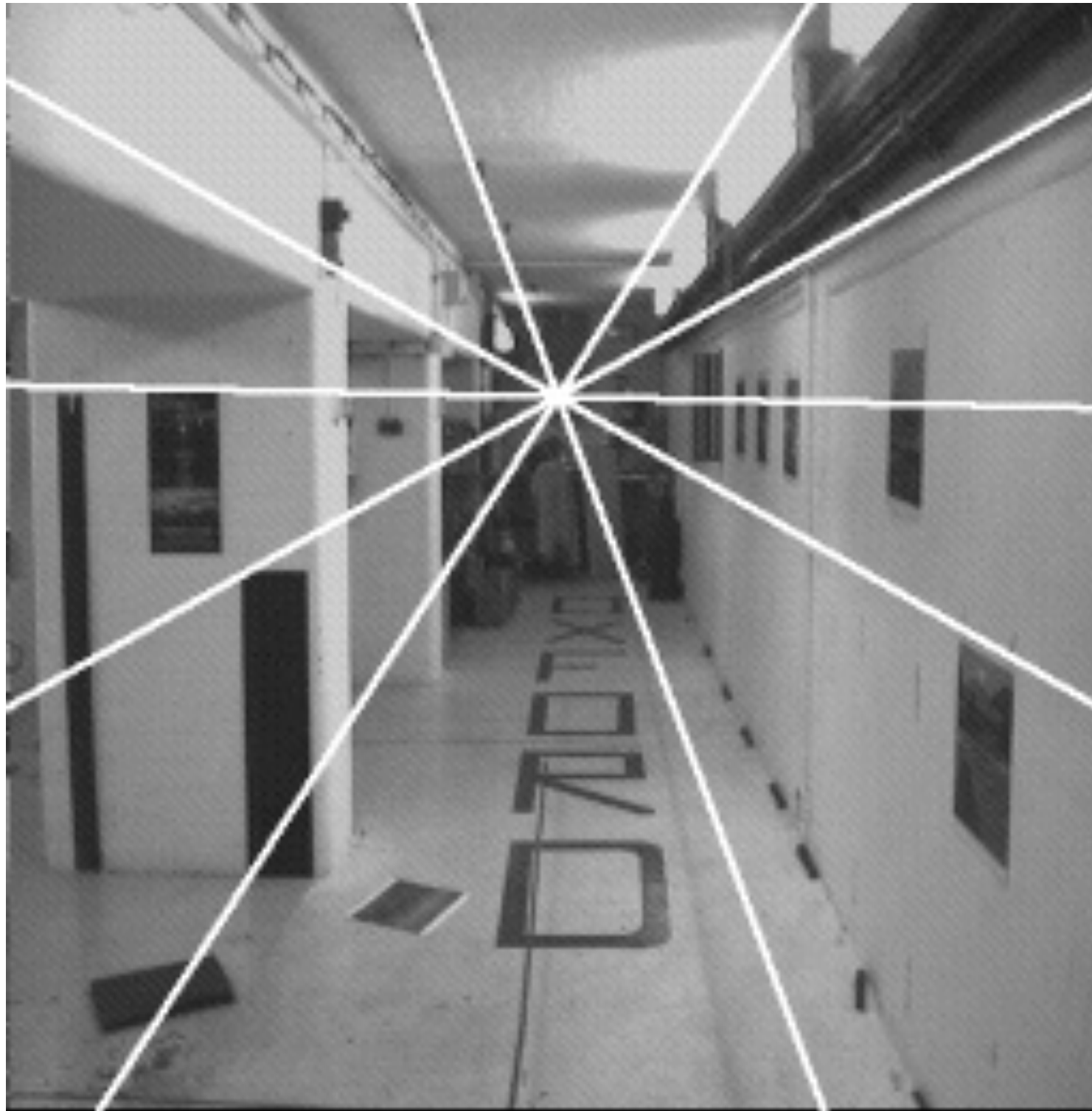
$$F = [e']_{\times}$$

$$e' = K \begin{bmatrix} I & t \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = Kt$$

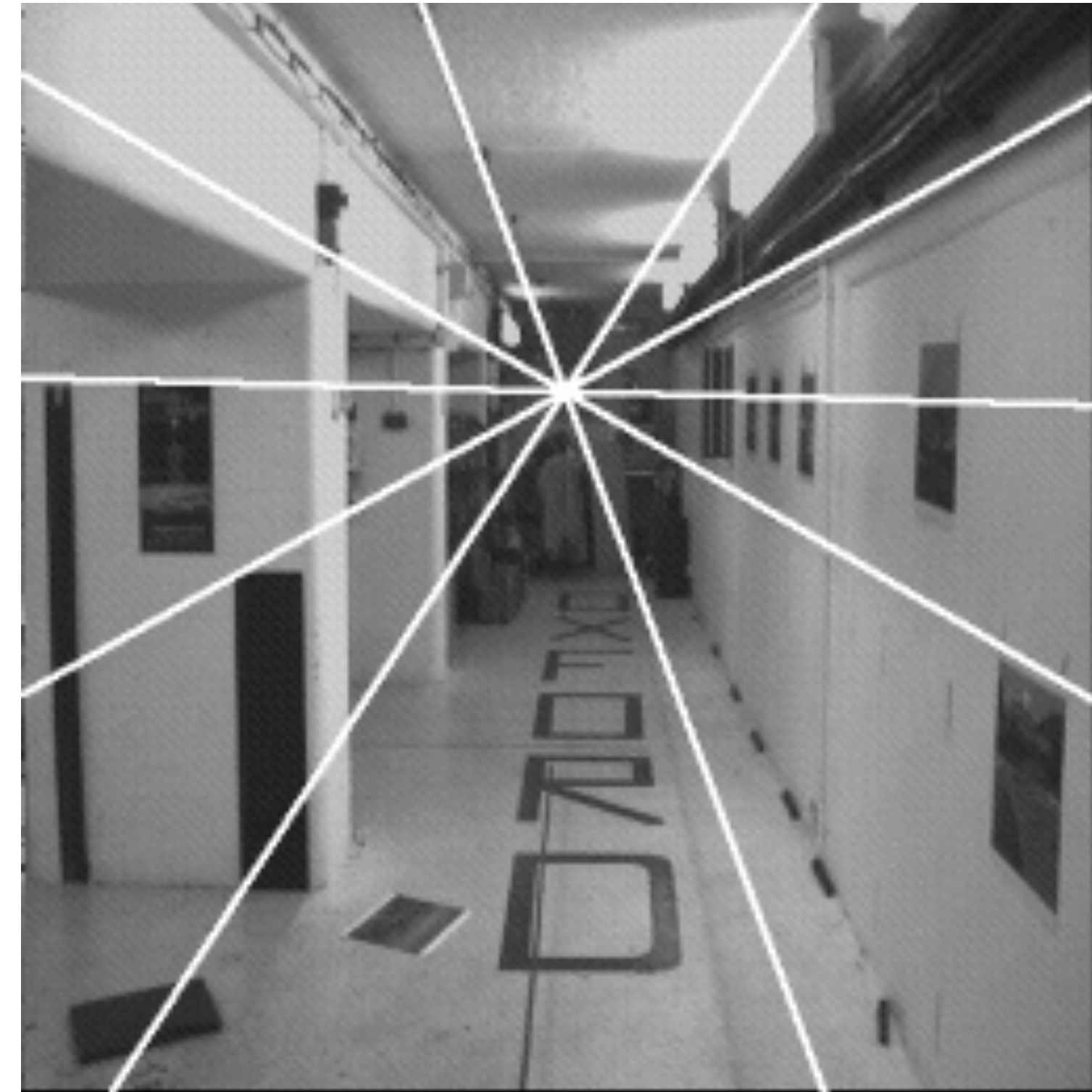
$$e = K \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} t \\ 1 \end{bmatrix} = Kt$$



Special Case: Translation (R=I, K=K')

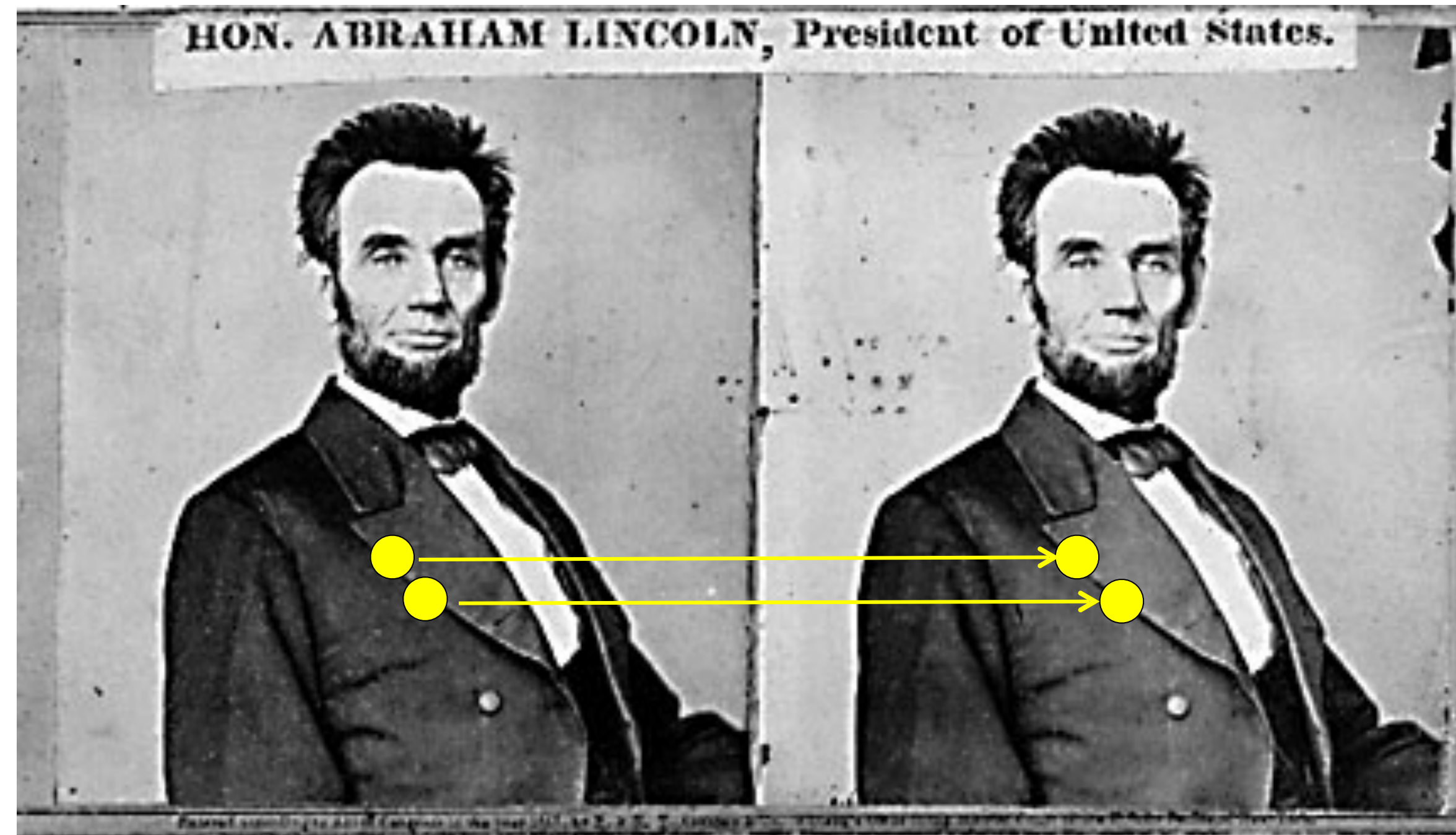


$$\mathbf{e}' = K \begin{bmatrix} I & \mathbf{t} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} = K\mathbf{t}$$



$$\mathbf{e} = K \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{t} \\ 1 \end{bmatrix} = K\mathbf{t}$$

Special Case: Binocular Stereo



Q: without any math, tell the coordinates of the epipole?

Essential Matrix: Relating Normalized Coordinates

$$\mathbf{x}'^T \mathbf{F} \mathbf{x} = 0$$

$$\hat{\mathbf{x}}'^T \mathbf{E} \hat{\mathbf{x}} = 0$$

$$\hat{\mathbf{x}} = \mathbf{K}^{-1} \mathbf{x}, \quad \hat{\mathbf{x}}' = \mathbf{K}'^{-1} \mathbf{x}'$$

Just as $\mathbf{F}$ relates image coordinates, $\mathbf{E}$ relates ‘normalized’ coordinates

$$\mathbf{E} = \mathbf{K}'^T \mathbf{F} \mathbf{K}$$

Given two cameras related by $\mathbf{R}, \mathbf{t}$:

$$\mathbf{F} = \mathbf{K}'^{-T} [\mathbf{t}]_{\times} \mathbf{R} \mathbf{K}^{-1}$$

$$\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R}$$

Q: How many degrees of freedom?

Two-view Reconstruction



Input: Two images of a scene



Output: (Sparse) 3D
Reconstruction + Camera Poses

Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract
correspondences



Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract correspondences

Step 2: Compute F (uncalibrated case) or E (calibrated case)



Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract correspondences



Step 2: Compute F (uncalibrated case) or E (calibrated case)



Step 3: Recover projection matrices P, P'

Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract correspondences



Step 2: Compute F (uncalibrated case) or E (calibrated case)



Step 3: Recover projection matrices P, P'



Step 4: Compute 3D via triangulation

Two-view Reconstruction: A simplified approach



Step 1: Annotate/extract correspondences



Step 2: Compute F (uncalibrated case) or E (calibrated case)



Step 3: Recover projection matrices P, P'



Step 4: Compute 3D via triangulation

next lecture

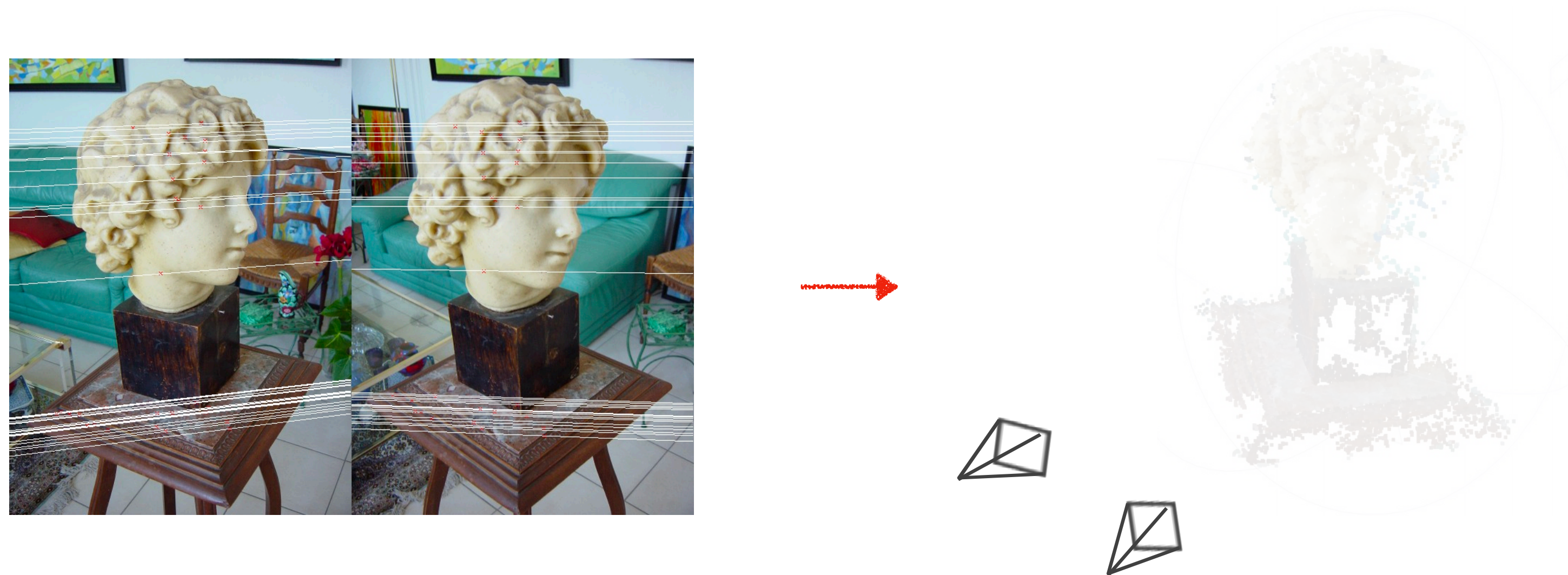
today's goal

next lecture

From Fundamental/Essential Matrix to Cameras

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Calibrated case: $E, (K, K') \rightarrow (\mathbf{P}, \mathbf{P}')$



Reminder: we know how to compute in the other direction!

$$F = [\mathbf{e}']_{\times} \mathbf{P}' \mathbf{P}^+ = \mathbf{K}'^{-\top} [\mathbf{t}]_{\times} \mathbf{R} \mathbf{K}^{-1}$$

$$\mathbf{E} = \mathbf{K}'^{\top} \mathbf{F} \mathbf{K} = [\mathbf{t}]_{\times} \mathbf{R}$$

Projective Ambiguity in Recovering Cameras from F

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Is it even possible to recover cameras?

Not uniquely!

Q: Can you think of some ambiguities?

Say F is the fundamental matrix between $(\mathbf{P}, \mathbf{P}')$

$$(\mathbf{P}, \mathbf{P}') \rightarrow F$$

Then, it is also the fundamental matrix relating $\mathbf{P}H$ and $\mathbf{P}'H$

$$(\mathbf{P}H, \mathbf{P}'H) \rightarrow F$$
$$\forall H \det(H) \neq 0$$

$$\forall \mathbf{X}, \quad \mathbf{X}^\top \mathbf{P}'^\top F \mathbf{P} \mathbf{X} = 0$$



reparametrize
 $\mathbf{X} = H\bar{\mathbf{X}}$
 $\det(H) \neq 0$

$$\forall \bar{\mathbf{X}}, \quad \bar{\mathbf{X}}^\top H^\top \mathbf{P}'^\top F \mathbf{P} H \bar{\mathbf{X}} = 0$$

Projective Ambiguity in Recovering Cameras from F

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Ambiguity upto a homography H:
 $(\mathbf{P}, \mathbf{P}')$ is a solution $\rightarrow (\mathbf{P}H, \mathbf{P}'H)$ is a solution

$$\begin{aligned} F &= [\mathbf{P}'\mathbf{C}]_{\times} \mathbf{P}' \mathbf{P}^{+} \\ &= [\mathbf{m}]_{\times} \mathbf{M} \end{aligned} \quad \begin{aligned} \mathbf{P} &= \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \\ \mathbf{P}' &= \begin{bmatrix} M & \mathbf{m} \end{bmatrix} \end{aligned}$$

(Canonical form)

Q: Why can we always convert to this form?

$$\mathbf{P}^* = \begin{bmatrix} \mathbf{P} \\ \mathbf{p}^* \end{bmatrix} \quad H = \mathbf{P}^{*-1}$$

pick any $\mathbf{p}^*$ s.t. $\mathbf{P}^*$ is non-singular

Note — canonical form conversion is not unique
(possible different choices of $\mathbf{p}^*$)

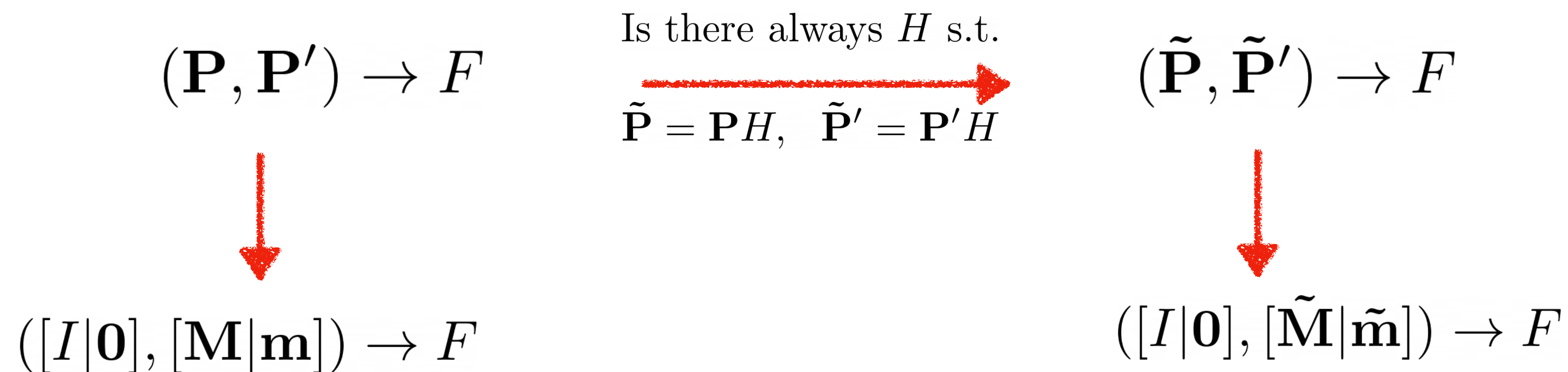
Projective Ambiguity in Recovering Cameras from F

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Ambiguity upto a homography H:
 $(\mathbf{P}, \mathbf{P}')$ is a solution $\rightarrow (\mathbf{P}H, \mathbf{P}'H)$ is a solution

Is this the only ambiguity?

Informal argument: how many degrees of freedom in $(\mathbf{P}, \mathbf{P}')$? how many resolved by F? how many in H?



Projective Ambiguity in Recovering Cameras from F

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Ambiguity upto a homography H:
 $(\mathbf{P}, \mathbf{P}')$ is a solution $\rightarrow (\mathbf{P}H, \mathbf{P}'H)$ is a solution

Is this the only ambiguity?

$$([I|\mathbf{0}], [\mathbf{M}|\mathbf{m}]) \xrightarrow[\mathbf{[m]_{\times}M = [m']_{\times}M'}{H \text{ given that}} ([I|\mathbf{0}], [\tilde{\mathbf{M}}|\tilde{\mathbf{m}}])$$

This constraint implies that for some $\mathbf{v}, k$ —

$$\tilde{\mathbf{m}} = k\mathbf{m} \quad \tilde{\mathbf{M}} = k^{-1}(\mathbf{M} + \mathbf{m}\mathbf{v}^T)$$

$$H = \begin{bmatrix} k^{-1}\mathbf{I} & \mathbf{0} \\ k^{-1}\mathbf{v}^T & k \end{bmatrix}$$

Projective Ambiguity in Recovering Cameras from F

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Ambiguity upto a homography H:
 $(\mathbf{P}, \mathbf{P}')$ is a solution $\rightarrow (\mathbf{P}H, \mathbf{P}'H)$ is a solution

Is this the only ambiguity?

Yes!

i.e. any two solutions are related by some
homography H

If the solutions are in the canonical form, they
are related by a homography of the form:

$$H = \begin{bmatrix} k^{-1} \mathbf{I} & \mathbf{0} \\ k^{-1} \mathbf{v}^T & k \end{bmatrix}$$

Recovering Cameras from F

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

But, what is a possible solution for
 $(\mathbf{P}, \mathbf{P}')$ given F ?

$$\mathbf{P} = [I \quad \mathbf{0}] \quad \mathbf{P}' = [M \quad \mathbf{m}] \quad \mathbf{M} = SF; \quad \mathbf{m} = \mathbf{e}'$$

$\mathbf{e}'$: left null-space of F S : any skew-symmetric matrix s.t. $\mathbf{P}'$ is rank 3

Proof: See that $\mathbf{P}'^T F \mathbf{P}$ is skew-symmetric

Possible choice: $S = [\mathbf{e}']_{\times}$

(as this enforces that columns of $\mathbf{M}$ are orthogonal to $\mathbf{m}$, ensuring rank=3)

Recovering Cameras from F

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

But, what is a possible solution for
 $(\mathbf{P}, \mathbf{P}')$ given F ?

$$\mathbf{P} = \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \quad \mathbf{P}' = \begin{bmatrix} [\mathbf{e}']_{\times} F & \mathbf{e}' \end{bmatrix}$$

Any other solution in the canonical form can be
computed via a homography of the form:

$$\mathbf{H} = \begin{bmatrix} k^{-1} \mathbf{I} & \mathbf{0} \\ k^{-1} \mathbf{v}^{\top} & k \end{bmatrix}$$

Recovering Cameras from E

Calibrated case: $E, (K, K') \rightarrow (\mathbf{P}, \mathbf{P}')$

Ambiguity upto a similarity H:
 $(\mathbf{P}, \mathbf{P}')$ is a solution $\rightarrow (\mathbf{P}H, \mathbf{P}'H)$ is a solution

$$\mathbf{P} = K \begin{bmatrix} I & \mathbf{0} \end{bmatrix}$$

$$\mathbf{P}' = K' \begin{bmatrix} R & \mathbf{t} \end{bmatrix}$$

(Canonical form)

Is this the only ambiguity?

Informal argument: 22 DoF in $(\mathbf{P}, \mathbf{P}')$.
5 + 5 + 5 resolved given F, K, K'.
How many in a similarity H?

Unlike recovery from F, the canonical form
solution from E is unique!
(well, almost)

Recovering Cameras from E

Calibrated case: $E, (K, K') \rightarrow (\mathbf{P}, \mathbf{P}')$

$$\mathbf{P} = K \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \quad \mathbf{P}' = K' \begin{bmatrix} R & \mathbf{t} \end{bmatrix}$$

$$R, \mathbf{t} \text{ s.t. } E = [\mathbf{t}]_{\times} R \quad \text{scale ambiguity in } \mathbf{t}$$
$$\|\mathbf{t}\| = 1$$

$$\begin{aligned} \begin{bmatrix} R & \mathbf{t} \end{bmatrix} &= [\mathbf{U}\mathbf{W}\mathbf{V}^{\top} \mid +\mathbf{u}_3] \text{ or } [\mathbf{U}\mathbf{W}\mathbf{V}^{\top} \mid -\mathbf{u}_3] \\ &\text{or } [\mathbf{U}\mathbf{W}^{\top}\mathbf{V}^{\top} \mid +\mathbf{u}_3] \text{ or } [\mathbf{U}\mathbf{W}^{\top}\mathbf{V}^{\top} \mid -\mathbf{u}_3] \end{aligned} \quad \mathbf{E} = \mathbf{U} \text{diag}(1, 1, 0) \mathbf{V}^{\top}$$
$$\mathbf{W} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(we'll skip derivation as it is more about linear algebra than geometry)

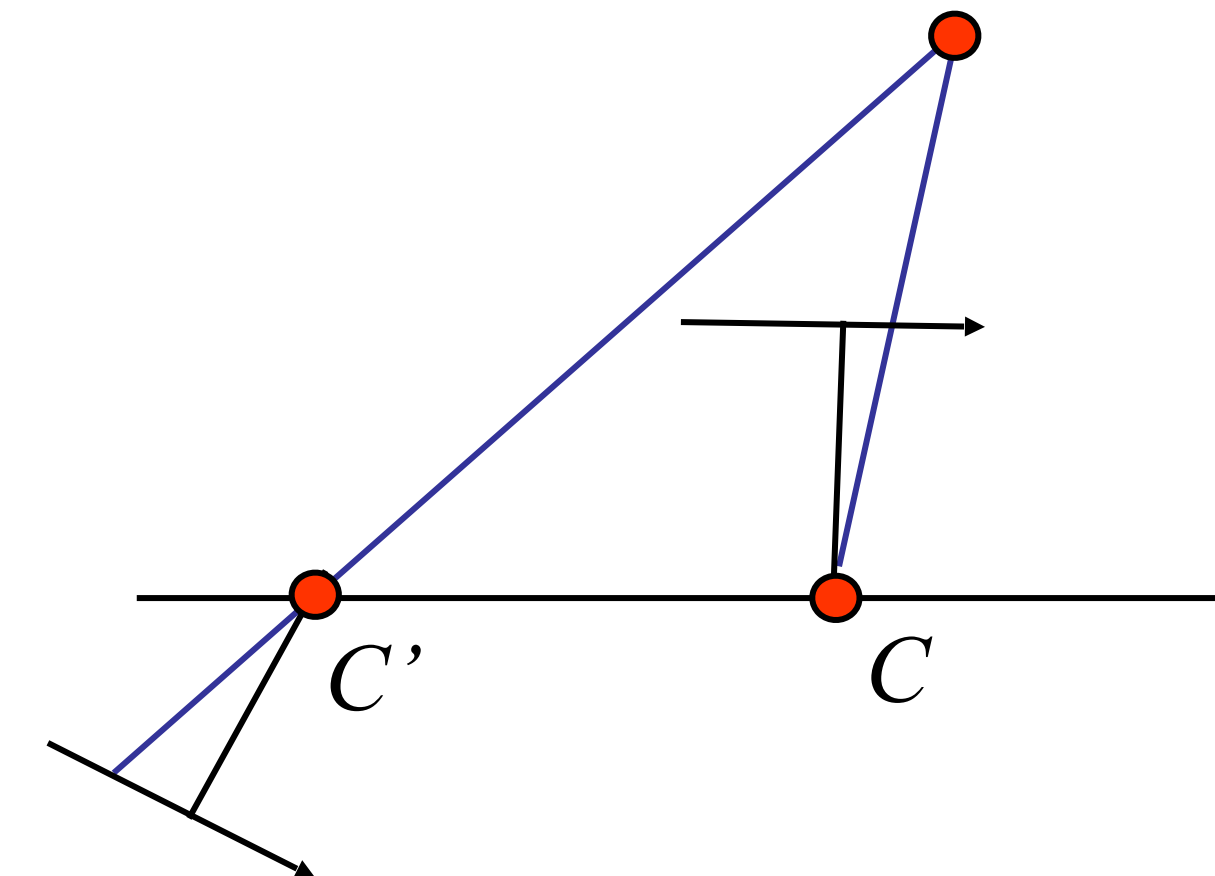
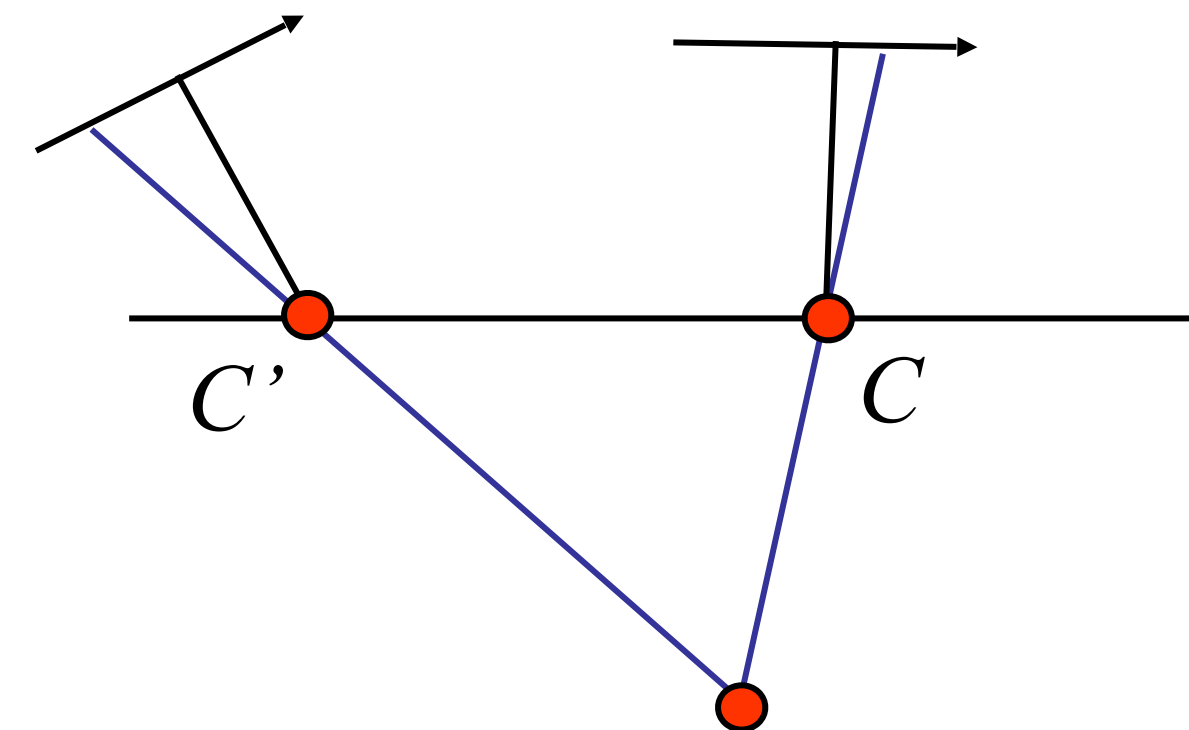
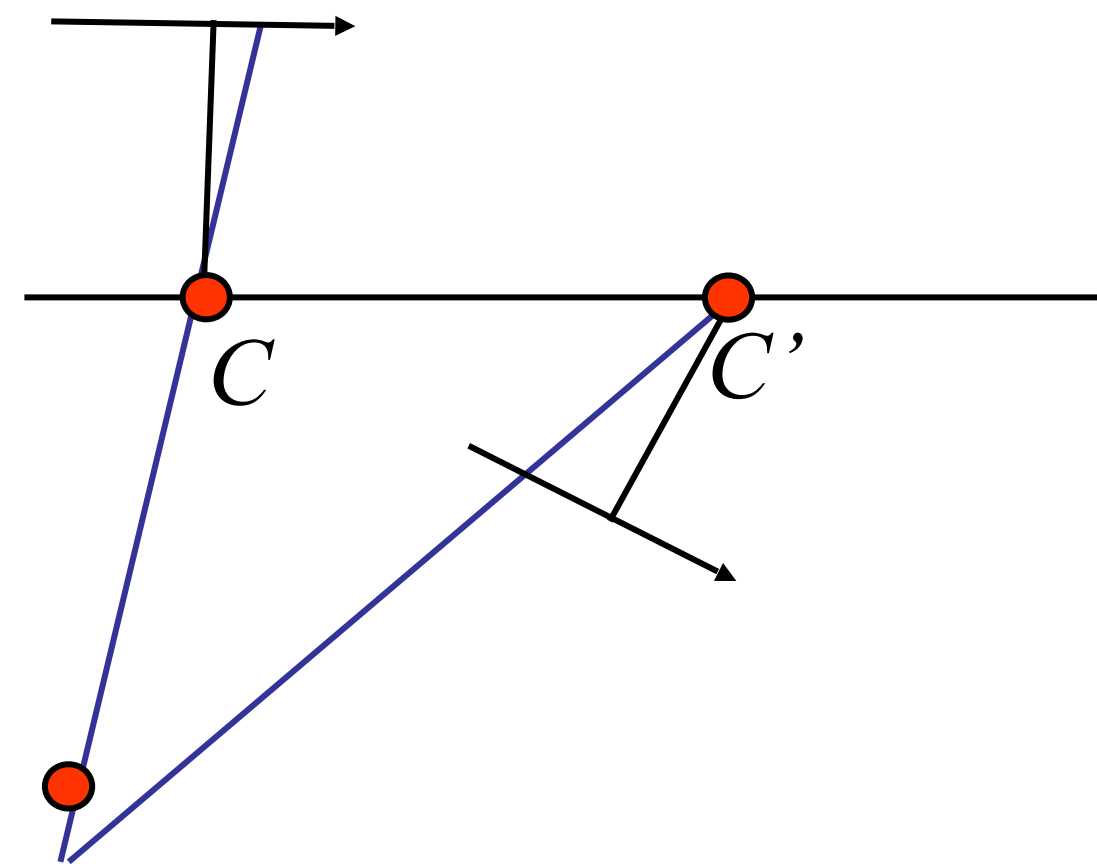
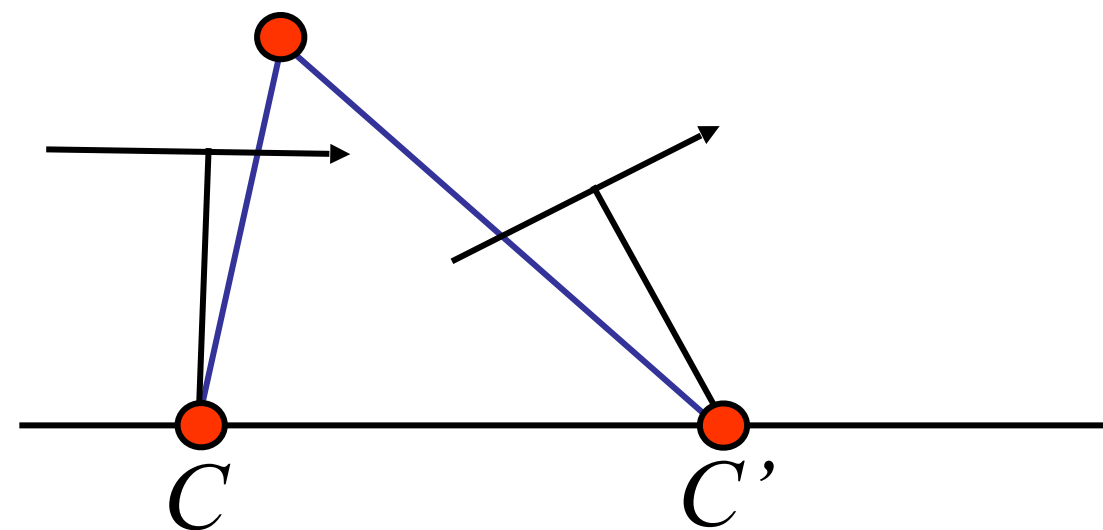
Recovering Cameras from E

Calibrated case: $E, (K, K') \rightarrow (\mathbf{P}, \mathbf{P}')$

$$\mathbf{P} = K \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \quad \mathbf{P}' = K' \begin{bmatrix} R & \mathbf{t} \end{bmatrix}$$

$$\begin{bmatrix} R & \mathbf{t} \end{bmatrix} = [\mathbf{U}\mathbf{W}\mathbf{V}^T \mid +\mathbf{u}_3] \text{ or } [\mathbf{U}\mathbf{W}\mathbf{V}^T \mid -\mathbf{u}_3]$$

$$\text{or } [\mathbf{U}\mathbf{W}^T\mathbf{V}^T \mid +\mathbf{u}_3] \text{ or } [\mathbf{U}\mathbf{W}^T\mathbf{V}^T \mid -\mathbf{u}_3]$$



Only one 'natural' solution (3D point in front of both cameras)
Can be identified via triangulation of one (or a few) point(s)

From Fundamental/Essential Matrix to Cameras

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Calibrated case: $E, (K, K') \rightarrow (\mathbf{P}, \mathbf{P}')$

$$\mathbf{P} = \begin{bmatrix} I & \mathbf{0} \end{bmatrix}$$

$$\mathbf{P} = K \begin{bmatrix} I & \mathbf{0} \end{bmatrix}$$

$$\mathbf{P}' = \begin{bmatrix} [\mathbf{e}']_{\times} F & \mathbf{e}' \end{bmatrix}$$

$$\mathbf{P}' = K' \begin{bmatrix} R & \mathbf{t} \end{bmatrix}$$

Any other solution in the canonical form can be computed via a homography of the form:

$$\mathbf{H} = \begin{bmatrix} k^{-1} \mathbf{I} & \mathbf{0} \\ k^{-1} \mathbf{v}^{\top} & k \end{bmatrix}$$

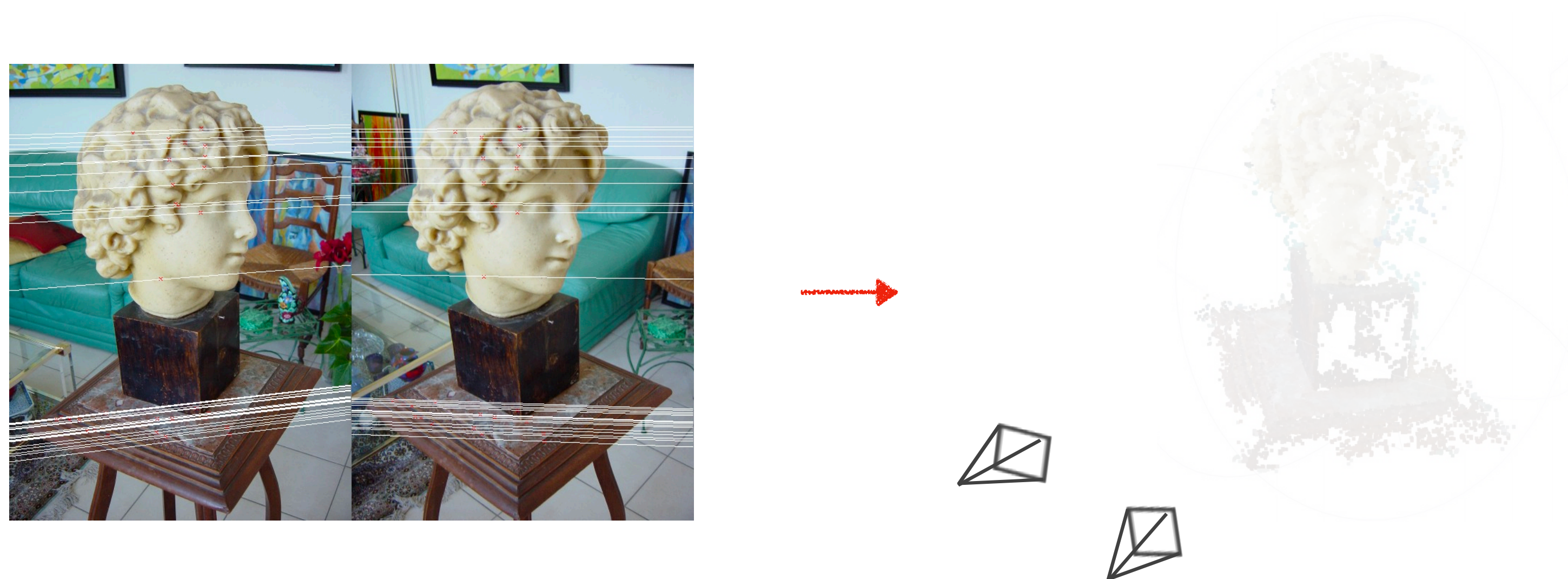
$$\begin{aligned} \begin{bmatrix} R & \mathbf{t} \end{bmatrix} &= [\mathbf{U}\mathbf{W}\mathbf{V}^{\top} \mid +\mathbf{u}_3] \text{ or } [\mathbf{U}\mathbf{W}\mathbf{V}^{\top} \mid -\mathbf{u}_3] \\ &\text{or } [\mathbf{U}\mathbf{W}^{\top}\mathbf{V}^{\top} \mid +\mathbf{u}_3] \text{ or } [\mathbf{U}\mathbf{W}^{\top}\mathbf{V}^{\top} \mid -\mathbf{u}_3] \end{aligned}$$

Any other solution in the canonical form stems from a scaling of $\mathbf{t}$

From Fundamental/Essential Matrix to Cameras

Uncalibrated case: $F \rightarrow (\mathbf{P}, \mathbf{P}')$

Calibrated case: $E, (K, K') \rightarrow (\mathbf{P}, \mathbf{P}')$



Questions?